

Information Gathering: Unobtrusive Methods

LEARNING OBJECTIVES

Once you have mastered the material in this chapter, you will be able to:

1. Recognize the value of unobtrusive methods of information gathering.
2. Understand the concept of sampling for human information requirements analysis.
3. Construct useful samples of people, documents, and events for determining human information requirements.
4. Interpret managers' and customers' messages, interviews, and communications using text analytics.
5. Create an analyst's playscript to observe decision makers' activities.
6. Apply the STROBE technique to observe and interpret a decision maker's environment and interaction with technologies.

Just by being present in an organization, a systems analyst changes it. However, unobtrusive methods such as sampling, investigating, and observing a decision maker's behavior and interacting with his or her physical environment are less disruptive than other ways of eliciting human information requirements. Text analytics is software that can analyze qualitative data from any unstructured written source. Interviews, written reports, or customers' communication collected through wikis, blogs, chat rooms, and other social networking sites provide analysts and decision makers with insights about the organization. Unobtrusive methods are insufficient information-gathering methods when used alone. Rather, they should be used in conjunction with one or more interactive methods such as those studied in the previous chapter. This is called a *multiple methods approach*. Using both interactive and unobtrusive methods in approaching an organization is a wise practice that will result in a more complete picture of human information requirements.

Sampling

Sampling is the process of systematically selecting representative elements of a population. When these selected elements are examined closely, analysis of the sample will reveal useful information about the population as a whole.

A systems analyst has to make decisions on two key issues. First, people in the organization have generated many reports, forms, output documents, memos, and websites. Which of these should the systems analyst pay attention to, and which can the systems analyst safely pass over?

Second, a great many employees may be affected by the proposed information system. Which people should the systems analyst interview, seek information from via questionnaires, or observe in the process of carrying out their decision-making roles?

The Need for Sampling

There are many reasons a systems analyst would want to select either a representative sample of data to examine or representative people to interview, question, or observe. They include:

1. Containing costs
2. Speeding up data gathering
3. Improving effectiveness
4. Reducing bias

Examining every scrap of paper, talking with everyone, and reading every web page from the organization is far too costly for the systems analyst. Copying reports, asking employees for valuable time, and duplicating unnecessary surveys would result in much needless expense. Sampling helps accelerate the process by gathering selected data rather than all data for the entire population. In addition, the systems analyst is spared the burden of analyzing data from the entire population.

Effectiveness in data gathering is an important consideration as well. Sampling can help improve effectiveness if information that is more accurate can be obtained. Such sampling is accomplished, for example, by talking to fewer employees but asking them more detailed questions. In addition, if fewer people are interviewed, the systems analyst can afford the time to follow up on missing or incomplete data, thus improving the effectiveness of data gathering.

Finally, data-gathering bias can be reduced by sampling. When the systems analyst interviews an executive of the corporation, for example, the executive is involved with the project because this person has already given a certain amount of time to the project and would like it to succeed. When the systems analyst asks for an opinion about a permanent feature of the installed information system, the executive interviewed may provide a biased evaluation because there is little possibility of changing it.

Sampling Design

A systems analyst must follow four steps to design a good sample:

1. Determine the data to be collected or described.
2. Determine the population to be sampled.
3. Choose the type of sample.
4. Decide on the sample size.

These steps are described in detail in the following subsections.

DETERMINING THE DATA TO BE COLLECTED OR DESCRIBED A systems analyst needs a realistic plan about what will be done with the data collected. If irrelevant data are gathered, then time and money are wasted in the collection, storage, and analysis of useless data.

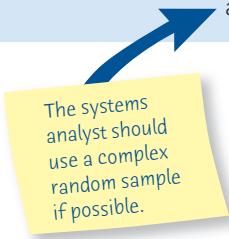
The duties and responsibilities of the systems analyst at this point are to identify the variables, attributes, and associated data items that need to be gathered in the sample. The objectives of the study must be considered, as well as the type of data-gathering method (for example, investigation, interviews, questionnaires, and observation) to be used. The kinds of information sought when using each of these methods are discussed in more detail in this and subsequent chapters.

DETERMINING THE POPULATION TO BE SAMPLED Next, a systems analyst must determine the size of the population. In the case of hard data, the systems analyst needs to decide, for example, whether

FIGURE 5.1

Four main types of samples a systems analyst has available.

	Not Based on Probability	Based on Probability
Sample elements are selected directly without restrictions	Convenience	Simple random
Sample elements are selected according to specific criteria	Purposive	Complex random (systematic, stratified, and cluster)



data from the last two months are sufficient, or if an entire year's worth of reports are needed for analysis.

Similarly, when deciding whom to interview, the systems analyst has to determine whether the population should include only one level in the organization or all the levels. Perhaps the analyst will need to go outside the system and include the reactions of customers, vendors, suppliers, or competitors as well. These decisions are explored in more detail in upcoming sections.

CHOOSING THE TYPE OF SAMPLE A systems analyst can use one of four main types of samples, as shown in Figure 5.1. They are convenience, purposive, simple random, and complex random. Convenience samples are unrestricted and nonprobability samples. A sample could be called a convenience sample if, for example, the systems analyst posts a notice on the company's intranet asking for everyone interested in working with the new sales performance reports to come to a meeting Tuesday morning. Obviously, this sample is the easiest to arrange, but it is also the least reliable. A purposive sample is based on judgment. The systems analyst chooses a group of individuals who appear knowledgeable and who are interested in the new information system. Although this sample is based on criteria—knowledge about and interest in the new system—purposive sampling is also a nonprobability sample and is only moderately reliable. If the analyst chooses a simple random sample, the sample must come from a numbered list of the population to ensure that each document or person in the population has an equal chance of being selected. This step often is not practical, especially when sampling involves documents and reports. Complex random samples that are most appropriate for a systems analyst are (1) systematic sampling, (2) stratified sampling, and (3) cluster sampling.

In the simplest method of probability sampling, systematic sampling, the systems analyst would, for example, choose to interview every k th person on a list of company employees. This method has certain disadvantages, however. The analyst would not want to use it to select every k th day for a sample because of the potential periodicity problem. Furthermore, a systems analyst would not use this approach if the list were ordered (for example, a list of banks, from the smallest to the largest) because bias would be introduced.

Stratified samples are perhaps the most important tool for a systems analyst. Stratification is the process of identifying subpopulations, or strata, and then selecting objects or people for sampling in these subpopulations. Stratification is often essential if the systems analyst is to gather data efficiently. For example, if the analyst seeks opinions from a wide range of employees on different levels of the organization, systematic sampling would select a disproportionate number of employees from the operational control level. A stratified sample compensates for this. Stratification also is called for when a systems analyst wants to use different methods to collect data from different subgroups. For example, the analyst may use a survey to gather data from middle managers but use personal interviews to gather similar data from executives.

Sometimes a systems analyst must select a group of people or documents to study. This process is referred to as *cluster sampling*. Suppose an organization has 20 help desks scattered

across the country. The analyst may select one or two of these help desks, under the assumption that they are typical of all the help desks.

DECIDING ON THE SAMPLE SIZE If everyone in the population viewed the world the same way or if each of the documents in a population contained exactly the same information as every other document, a sample size of one would be sufficient. Of course, this is not the case, so it is necessary to set a sample size greater than one but less than the size of the population itself.

It is important to remember that the absolute number is more important in sampling than the percentage of the population. We can obtain satisfactory results sampling 20 people in 200 or 20 people in 2,000,000.

The Sample Size Decision

The sample size often depends on the cost involved or the time required by the systems analyst, or even the time available from people in the organization. This subsection provides guidelines for determining the required sample size under ideal conditions—for example, to determine what percentage of input forms contain errors or what proportion of people to interview.

A systems analyst follows seven steps, some of which involve subjective judgments, to determine the required sample size:

1. Determine the attribute (in this case, the type of errors to look for).
2. Locate the database or reports in which the attribute can be found.
3. Examine the attribute. Estimate p , the proportion of the population having the attribute.
4. Make the subjective decision regarding the acceptable interval estimate, i .
5. Choose the confidence level and look up the confidence coefficient (z value) in a table.
6. Calculate σ_p , the standard error of the proportion, as follows:

$$\sigma_p = \frac{i}{z}$$

7. Determine the necessary sample size, n , using the following formula:

$$n = \frac{p(1-p)}{\sigma_p^2} + 1$$

The first step, of course, is to determine which attribute to sample. Once this is done, find out where these data are stored, perhaps in a database, on a form, or in a report.

It is important to estimate p , the proportion of the population having the attribute, to set the appropriate sample size. Many textbooks on systems analysis suggest using a heuristic of 0.25 for $p(1-p)$. This value almost always results in a sample size larger than necessary because 0.25 is the maximum value of $p(1-p)$, which occurs only when $p = 0.50$. When $p = 0.10$, as is more often the case, $p(1-p)$ becomes 0.09, resulting in a much smaller sample size.

Steps 4 and 5 are subjective decisions. The acceptable interval estimate of ± 0.10 means the analyst is willing to accept an error of no more than 0.10 in either direction from the actual proportion, p . The confidence level is the desired degree of certainty, such as 95 percent. Once the confidence level is chosen, the confidence coefficient (also called a z value) can be looked up in a table similar to the one found in this chapter.

Steps 6 and 7 complete the process by taking the parameters found or set in Steps 3 through 5 and entering them into two equations to eventually solve the required sample size.

EXAMPLE

The foregoing steps can best be illustrated by an example. Suppose the A. Sembly Company, a large manufacturer of shelving products, asks you to determine what percentage of orders contain errors. You agree to do this job and perform the following steps:

1. Determine that you will be looking for orders that contain mistakes in names, addresses, quantities, or model numbers.
2. Locate copies of order forms from the past six months.
3. Examine some of the order forms and conclude that only about 5 percent (0.05) contain errors.



CONSULTING OPPORTUNITY 5.1

Trapping a Sample

“Real or fake? Fake or real? Who would have thought it, even five years ago?” howls Sam Pelt, a furrier who owns stores in New York, Washington, D.C., Beverly Hills, and Copenhagen. Sylva Foxx, a systems analyst with her own consulting firm, is talking with Sam for the first time. Currently, P & P, Ltd. (which stands for Pelt and Pelt’s son) is using a networked computer that supports package software for a select customer mailing list, accounts payable and accounts receivable, and payroll.

Sam is interested in making some strategic decisions that will ultimately affect the purchasing of goods for his four fur stores. Although the computer might help, he believes other approaches should be considered as well.

Sam continues, “I think we should talk to all the customers when they come in the door. Get their opinions. You know, some of them are getting very upset about wearing fur from endangered species. They’re very environmentally minded. They prefer fake to real, if they can save a baby animal. Some even like fakes better, calling them ‘fun furs.’ And I can charge almost the same for a good look-alike.”

“It’s a very fuzzy proposition, though. If I get too far away from my suppliers of pelts, I may not get what I want when I need it. They see the fake fur people as worms, worse than moths! If I deal with them, the real fur men might not talk to me. They can be animals. On the other hand, I feel strange showing fakes in my stores. All these years, we’ve prided ourselves on having only the genuine article.”

Sam continues, in a nearly seamless monologue, “I want to talk to each and every employee too.”

Sylva glances at him furtively and begins to interrupt, “But that will take months, and purchasing may come apart at the seams unless they know soon what . . .”

Pelt interrupts, “I don’t care how long it takes if we get the right answers. But they have to be right. Not knowing how to solve this dilemma about fake furs is making me feel like a leopard without its spots.”

Sylva talks to Sam Pelt a bit longer and then ends the interview by saying, “I’ll talk it all over with the other analysts at the office and let you know what we come up with. I think we can outfox the other furriers if we use software to help us sample opinions rather than trapping unsuspecting customers into giving an opinion. But I’ll let you know what they say. This much is for sure: if we can sample and not talk to everybody before making a decision, every coat you sell will have a silver lining.”

As one of the systems analysts who is part of Sylva Foxx’s firm, suggest some ways that Sam Pelt can use software to adequately sample the opinions of his customers, store managers, buyers, and any others you feel will be instrumental in making the strategic decision regarding the stocking of fake furs in what has always been a real fur store. In a paragraph, suggest a type of sample for each group and justify it. The constraints you are subject to include the need to act quickly to remain competitive, the need to retain a low profile so competing furriers are unaware of your fact gathering, and the need to keep costs of data gathering to a reasonable level.

4. Make a subjective decision that the acceptable interval estimate will be ± 0.02 .
5. Choose a confidence level of 95 percent. Look up the confidence coefficient (z value) in Figure 5.2. The z value equals 1.96.
6. Calculate σ_p as follows:

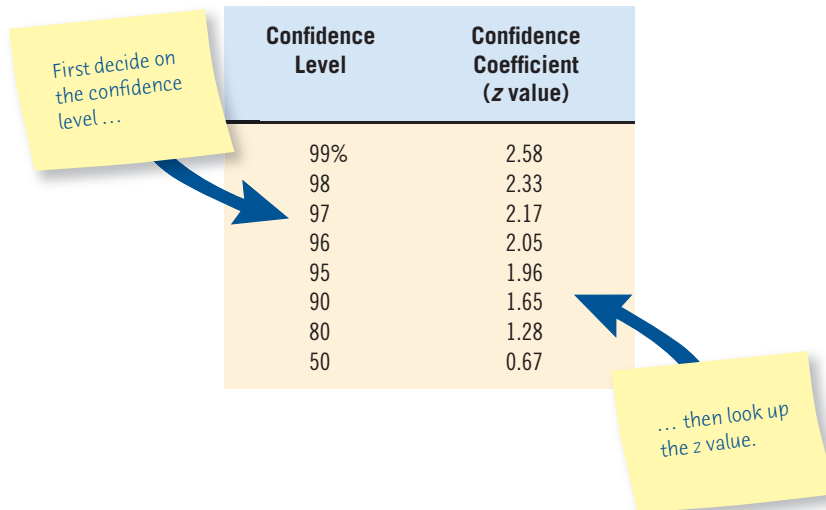
$$\sigma_p = \frac{i}{z} = \frac{0.02}{1.96} = 0.0102$$

7. Determine the necessary sample size, n , as follows:

$$n = \frac{p(1-p)}{\sigma_p^2} + 1 = \frac{0.05(0.95)}{(0.0102)(0.0102)} + 1 = 458$$

The conclusion, then, is to set the sample size at 458. Obviously, a greater confidence level or a smaller acceptable interval estimate would require a larger sample size. If we keep the acceptable interval estimate the same but increase the confidence level to 99 percent (with a z value of 2.58), the necessary sample size becomes 1,827, a figure much larger than the 458 we originally decided to sample.

DETERMINING SAMPLE SIZE WHEN INTERVIEWING There are no magic formulas to help a systems analyst set the sample size for interviewing. The overriding variable that determines how many people a systems analyst should interview in depth is the time an interview takes. A true in-depth interview and a follow-up interview are very time consuming for both the interviewer and the participant.



Confidence Level	Confidence Coefficient (z value)
99%	2.58
98	2.33
97	2.17
96	2.05
95	1.96
90	1.65
80	1.28
50	0.67

FIGURE 5.2

A systems analyst can use a table of area under a normal curve to look up a value once he or she decides on the confidence level.

A good rule of thumb is to interview at least three people at every level of the organization and at least one from each of the organization's functional areas (as described in Chapter 2) who will work directly with a new or updated system. Remember also that it is not necessary to interview more people just because it is a larger organization. If the stratified sample is done properly, a small number of people will adequately represent the entire organization.

Analyzing Quantitative Documents

Investigation is the act of discovery and analysis of data. When investigating evidence in an organization, an analyst acts like Sherlock Holmes, the fabled detective from 221B Baker Street.

As a systems analyst works to understand users, their organization, and its information requirements, it becomes important to examine different types of hard data that offer information unavailable through any other method of data gathering. Hard data reveal where the organization has been and where its members believe it is going. To piece together an accurate picture, the analyst needs to examine both quantitative and qualitative hard data.

Many quantitative documents are available for interpretation in any business, and they include reports used for decision making, performance reports, records, and a variety of forms. All of these documents are targeted for specific purposes and audiences.

Systematically Examining Quantitative Documents

A systems analyst needs to obtain some of the documents used in running the business. These documents often are paper reports regarding the status of inventory, sales, production, and employee and customer satisfaction. Many of these reports are not complex, and they can provide feedback for quick action. For example, a sales report may summarize the amount sold and the type of sales. In addition, sales reports might include graphical output comparing revenue and income over a set number of periods. Such reports enable the decision maker to spot trends easily.

Production reports include recent costs, current inventory, labor, and plant information. Service reports include the delivery of services and reports on employee performance and customer satisfaction. In addition to using these key reports, decision makers use many summary reports to provide background information, spot exceptions to normal occurrences, and afford strategic overviews of organization plans. The following reports, records, and forms deserve special attention.

PERFORMANCE REPORTS Most performance reports take the general form of actual versus intended performance. One important function of performance reports is to assess the size of the gap between actual and intended performance. It is important to be able to determine whether that gap is widening or narrowing as an overall trend in whatever performance is being measured. Figure 5.3 shows a clear improvement in sales performance over two to three months. The analyst will want to note whether a performance measurement is available and adequate for key organization areas.

RECORDS Records provide periodic updates of what is occurring in the business. If the record is updated in a timely fashion by a careful recorder, it can provide much useful information to a



CONSULTING OPPORTUNITY 5.2

A Rose by Any Other Name . . . Or Quality, Not Quantities

“I think we have everything we need. I’ve sampled financial statements, sales figures for each branch, waste for each shop—we have it all. With all these numbers, we should be able to figure out how to keep Fields in the green, or at least at the forefront of the flower business. We can even show Seymour Fields himself how his new computer system can make it all happen,” says Rod Golden, a junior systems analyst working for a medium-sized consulting group.

The firm, under the supervision of its head systems analyst Clay Potts, has been working on a systems project for the entire chain of 15 successful florist shops and indoor floral markets called Fields. Each of three Midwestern cities has five Fields outlets.

“Although it’s just a budding enterprise now, eventually we want to grow with offshoots to half a dozen states,” says Seymour Fields, the owner. “I want to reap the benefits of all the happiness we’ve sown so far. I think we can do it by playing my hunches about what is the best time to purchase flowers at each European market we buy from, and then we should prune back our purchases.”

“Over the past three years, I’ve written lots of memos to our managers about this plan. They’ve written some good ones back too. I think we’re ready to stake out some territory on this soon,” continues Seymour, painting a rosy picture of Fields’s future.

“I agree,” says Rod. “When I come back from my analysis of these figures,” he says, indicating a large stack of material he has unearthed from Fields field offices, “we’ll be able to deliver.”

Three weeks later, Rod returns to Clay with wilting confidence. “I don’t know what to make of all this. I can’t seem to get at what’s causing the company’s growth, or how it’s managed. They’ve been expanding, but I’ve been through all the figures, and nothing really seems to make sense yet.”

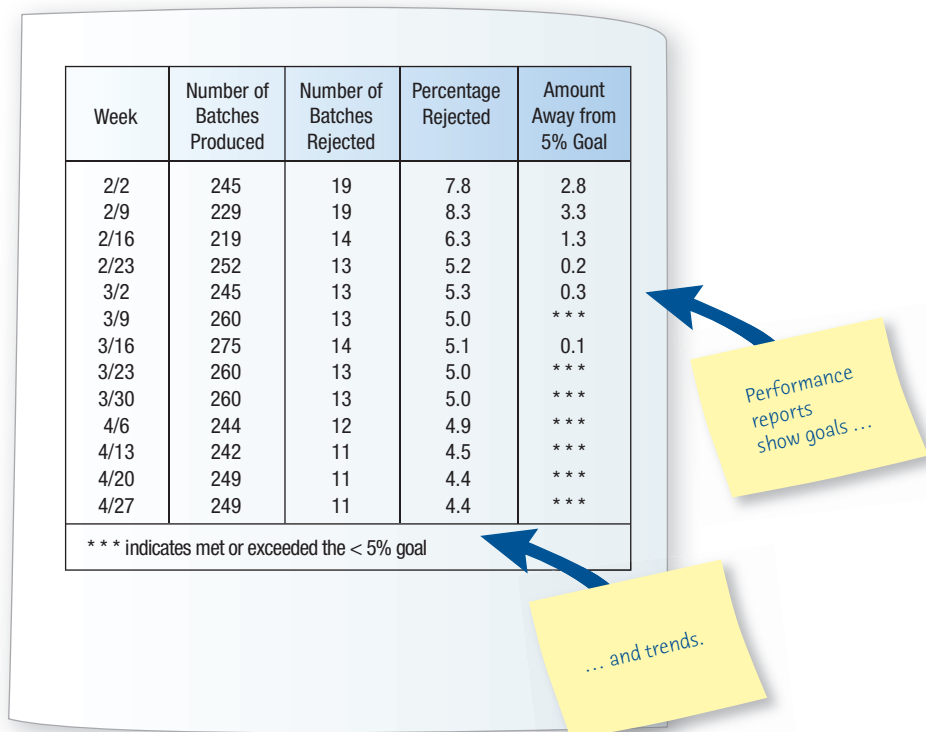
Clay listens empathetically, and then says, “You’ve given me a germ of an idea. What we need is some cross-pollination, a breath of fresh air. We need to dig a little deeper. Did you examine anything but their bottom line?”

Rod looks startled and replies, “No, I—uh—what do you mean?”

How can Clay Potts tactfully explain to Rod Golden that examination of qualitative as well as quantitative documents could be important in delivering an accurate assessment of Fields’s potential to be a more fruitful enterprise? In a paragraph, recommend some specific documents that should be read. List the specific steps Rod should follow in evaluating qualitative documents obtained from Fields. Write a paragraph to explain how qualitative documents help in presenting an overall account of Fields’s success.

FIGURE 5.3

A performance report showing improvement.



Check for errors.

Look for opportunities for improvement in design.

PROJ. NAME OAK. FC # 562 KEY SIGNATURE _____

RENT POTENTIAL				1175/0	81299	DEPOSIT POTENTIAL				PRORATE				15 ⁰⁰ 121 ³²		
Base Rent	Refrigerator	Furniture	A/C	Util.	HMSR	T.V.	Maid	Total Rent	Security	Cleaning	31175/0	81299	31700 Tax	Days	Daily Rate	Totals
855		55						910			H/S dep.	H/S rent		4	30 ³³ 4.30	5 ²⁰ 910 ³⁹
									200	115	Deposits				31 ⁶³	340

PAYMENT RECORD: Tot. 31175/0 + 81299 + Rent = 910 TOTAL INITIAL PAYMENT REQUIRED: 1430.⁵²

Memo Only	Date Due	Date Paid	Receipt Number	Paid to Noon	Total Rent	Security	Cleaning	31700 Tax	31175/0	81299 Dates/Amt.	Other Descr./Amt.	Amount Paid	Balance Due
TV 10/3 MO!	8/28	8/28	106642	9/30	1031. ³²	202	115	44. ²⁰	25		414. ⁸²	15	1430. ⁵²
	10/1	10/3	107503	10/31	910							910	0
	11/1	11/1	10935	11/16	485. ²⁸							485. ²⁸	0
C1H/S9-16	11/17	11/8	11200	11/23	212. ³¹							212. ³¹	0
Bill 1 MO	11/24												
Prorated													
H/S should be created toward refund deposit													

Orig. Move-in Date 8-28 d same Exp. _____ x # 1

BLDG. # _____ NAME Kendall 1st

Observe the number and type of transactions.

Watch for places the computer can simplify the work.

FIGURE 5.4

A manually completed payment record.

systems analyst. Figure 5.4 is a manually completed payment record for an apartment rental. An analyst can inspect a record in different ways, many of which are indicative of their usability:

1. Checking for errors in amounts and totals
2. Looking for opportunities for improving the recording form design
3. Observing the number and type of transactions
4. Watching for instances in which the computer can simplify the work through calculations and other data manipulation

DATA CAPTURE FORMS Before you set out to change the information flows in an organization, you need to be able to understand the system that is currently in place. Check with managers to determine which forms are available online to customers as well as those that are exclusively used internally by employees. You or one of your team members may want to download, collect, and catalog a blank copy of each form (official or unofficial) that is in use. (Sometimes businesses have a person already charged with forms management, who would be your first source for actual forms in use.) Remember to print out any Web-based forms that require users to print them. Alternatively, electronic versions that can be submitted via the Web or email can be identified and stored in a database for later inspection.

Blank forms, along with their instructions for completion and distribution, can be compared with filled-in forms to see whether any data items are consistently left blank on the forms; whether the people who are supposed to receive the forms actually do get them; and if they follow standard

procedures for using, storing, and discarding them. When creating a catalog of forms to help you understand the information flow currently in use in the business, follow these steps:

1. Collect examples of all the forms in use, whether officially sanctioned by the business or not (official versus bootleg forms).
2. Note the type of form (printed in-house, handwritten, computer-generated in-house, online forms, Web fill-in forms, printed externally and purchased, and so forth).
3. Document the intended distribution pattern.
4. Compare the intended distribution pattern with who actually receives the form.

Although this procedure is time consuming, it is useful. Another approach is to sample data capture forms that have already been completed. Remember to check databases that store consumer data when sampling input from ecommerce transactions. An analyst must keep in mind many particular questions, as illustrated in Figure 5.5. They include the following aspects of human-computer interaction (HCI) relating to usability, aesthetics, and usefulness:

1. Is the form filled out in its entirety? If not, what items have been omitted, and are they consistently omitted? Why?

FIGURE 5.5

Questions to ask about official and bootleg forms that are already filled out.

Farmfresh
Reorder of Shorted Dairy Products

Date _____ Store Name _____ Store Number _____

Item Requested	Cases	Item Requested	Cases
Milk (1/2 gals.)		Milk (quarts)	
Whole	_____	Whole	_____
2%	_____	2%	_____
1%	_____	1%	_____
Skim	_____	Skim	_____
Buttermilk	_____	Buttermilk	_____
Chocolate	_____	Chocolate	_____
Yogurt			
Plain	_____	Pineapple	_____
Vanilla	_____	Dutch Apple	_____
Peach	_____	Banana	_____
Blueberry	_____	Mixed Fruit	_____
Boysenberry	_____	Raspberry	_____
Strawberry	_____	Lemon	_____
Ice Cream			
Deluxe Pints	_____	Deluxe Quarts	_____
Deluxe 1/2 Gallons	_____	Premium Pints	_____
Skinny Minnies	_____	Premium Quarts	_____
Requested by (employee number) _____		Total Cases Ordered _____	
Reason for Shortage _____			
Driver Number _____		Route Number _____	

Requested by (employee number) _____ Total Cases Ordered _____

Reason for Shortage _____

Driver Number _____ Route Number _____

Official form can overwhelm people by asking for too much information.

There may be no logical order to the form.

Is the total really needed?

"Bootleg" forms arise to simplify the problem.

Store _____ Date _____ Driver _____

Product shorted	Cases needed
_____	_____
_____	_____
_____	_____
_____	_____

Dairy manager's initials _____

2. Are some forms never used? Why? (Check the design and appropriateness of each form for its purported function.)
3. Are all copies of forms circulated to the proper people or filed appropriately? If not, why not? Can people who must access online forms do so? Check on permissions needed and whether links to forms are functioning properly.
4. If a paper form is offered as an alternative to a Web-based form, compare the completion rates for both.
5. Are “unofficial” or bootleg forms being used on a regular basis? (This may indicate a problem in standard procedures or may indicate political battles in the organization over who is controlling the type of input and who enters data.)

Analyzing Qualitative Documents

Qualitative documents include written email messages, memos, signs on bulletin boards and in work areas, web pages, procedure manuals, and policy handbooks. Many of these documents are rich in details that reveal the expectations for behavior of others that their writers’ hold and the ways in which users expect to interact with information technologies.

Systematically Examining Qualitative Documents

Although many systems analysts are apprehensive about analyzing qualitative documents, they need not be. Several guidelines can help analysts take a systematic approach to this sort of analysis. Many of these relate to the affective, emotional, and motivational aspects of HCI, as well as to interpersonal relationships in the organization:

1. Examine documents for key or guiding metaphors.
2. Look for insiders versus outsiders or an “us against them” mentality.
3. List terms that characterize good or evil and appear repeatedly in documents.
4. Look for the use of meaningful messages and graphics posted on common areas or on web pages.
5. Recognize a sense of humor, if present.

It is important to examine documents for key or guiding metaphors because language shapes behavior; thus, the metaphors we employ are critical. For example, an organization that discusses employees as “part of a great machine” or “cogs in a wheel” might be taking a mechanistic view of the organization. Notice that the guiding metaphor in the memo in Figure 5.6 is “We’re one big happy family.” An analyst can use this information to predict the kinds of metaphors that will be persuasive in the organization. Obviously, if one department is battling another, it may be impossible to gain any cooperation on a systems project until the politics are resolved in a satisfactory manner. Assessing the use of humor provides a quick and accurate barometer of many

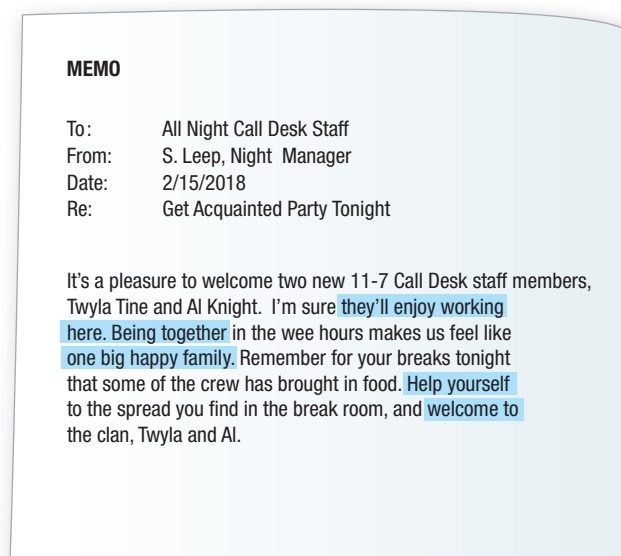


FIGURE 5.6

Analysis of memos provides insight into the metaphors that guide the organization’s thinking.



HYPERCASE EXPERIENCE 5.1

“We’re glad you find MRE an interesting place to consult. According to the grapevine, you’ve been busy exploring the home office. I know, there’s so much going on. We find it hard to keep track of everything ourselves. One thing we’ve made sure of over the years is that we try to use the methods that we believe in. Have you seen any of our reports? How about the data that were collected on one of Mr. Evan’s questionnaires? He seems to favor questionnaires over any other method. Some people resent them, but I think you can learn a lot from the results. Some people have been good about cooperating on these projects. Have you met Kathy Blandford yet?”

HYPERCASE Questions

1. Use clues from the case to evaluate the Training Unit’s computer experience and its staff’s feeling about the PSRS. What do you think the consensus is in the Training Unit toward a computerized project tracking system?
2. What reports and statements are generated by the Training Unit during project development? List each, with a brief description.
3. According to the interview results, what are the problems with the present project tracking system in the Training Unit?
4. Describe the “project management conflict” at MRE. Who is involved? Why is there a conflict?
5. How does the Management Systems Unit keep track of project progress? Briefly describe the method or system.

HCI, interpersonal, and organizational variables, including which subculture a person belongs to and what kind of morale exists.

MEMOS Along with the five preceding guidelines, an analyst should consider who sends memos (check email threads) and who receives them. Typically, most information flows downward and horizontally rather than upward in organizations, and extensive email systems mean messages are sent to many work groups and individuals. Memos reveal a lively, continuing dialogue in the organization. Analysis of memo content provides the analyst with a clear idea of the values, attitudes, and beliefs of organization members.

SIGNS OR POSTERS ON BULLETIN BOARDS OR IN WORK AREAS Although signs may seem incidental to what is happening in an organization, they serve as subtle reinforcers of values to those who read them. Slogans posted such as “Quality Is Forever” or “Safety First” give an analyst a sense of the official organizational culture.

CORPORATE WEBSITES An analyst should view websites used for business-to-consumer (B2C) ecommerce as well as those used for business-to-business (B2B) transactions. Examine the contents for metaphors, humor, use of design features (such as color, graphics, animation, and hyperlinks), and the meaning and clarity of any messages provided. Think about the website from three dimensions: technical, aesthetic, and managerial. Are there discrepancies between the stated goals of the organization and what is presented to the intended viewer? How much customization of the website is available for each user? How much personalization of the website is possible? If the analyst is not designing ecommerce sites for the organization, how does what the analyst sees on its website affect the systems being investigating? Remember to note the level of interactivity of the website or sites, the accessibility of the messages, and the security level. Many of these metrics are available from the company’s Internet provider and they can provide a different look at the company. Analysts should ask for access to this type of data if it is thought to be relevant to the systems being proposed.

MANUALS Other qualitative documents an analyst should examine are organizational manuals, including manuals for computer operating procedures and online manuals. Analyze manuals following the five guidelines spelled out previously. Remember that manuals present the “ideal,” the way machines and people are expected to behave. It is important to recall that printed manuals are rarely kept current and are sometimes relegated to shelves, unused.

POLICY HANDBOOKS The last type of qualitative document we consider is the policy handbook. Although these documents typically cover broad areas of employee and corporate behavior, a

systems analyst is primarily concerned with those that address policies about computer services, use, access, security, and charges. Examining policies allows a systems analyst to gain awareness of the values, attitudes, and beliefs guiding the corporation.

Using Text Analytics

Text analytics is software that can analyze unstructured qualitative data from any source including transcripts of interviews, written reports, or customers' communication collected through email, wikis, blogs, chat rooms, and other social networking sites. These data provide analysts and decision makers with insights about the organization. You may have the opportunity to develop systems that help an organization's users interpret and understand the unstructured data in their companies. Unstructured, qualitative, or "soft" data are generated through blogs, chat rooms, questionnaires using open-ended questions, online discussions conducted on the Web, and exchanges occurring on social media such as Facebook, Twitter, and other Web-generated dialogues between customers and an organization. Text analytics can help organizational decision makers realize valuable insights from unstructured data into what customers are thinking about the organization, the values and actions of the company, as well as customer or vendor motivations for beginning, maintaining, improving, or discontinuing a relationship. (In contrast, you will learn in Chapter 13 that structured data is well organized and searchable by standard queries and algorithms through data analytics. An example is data stored in a relational database or data gathered from machine sensors. The largest portion of an organization's information comes from this type of analysis.)

Many large and small analytics software solutions are available, both proprietary and open source. If you have been active in helping an organization develop and implement an online presence, then incorporating text analytics into the new systems makes a lot of sense. As you design systems to support an online presence, text analytics should be included to help the company make use of its online presence.

Text analytics provide insights for an organization's members who want to have a rapid and visual yet decidedly qualitative approach to analyzing text data. Leximancer, a powerful text analytics software tool from an Australian company, offers a straightforward approach for which no setup is required. Rather, the role of the organizational user who will use the software is to contribute relevant knowledge about what is to be fed into the system so concepts and other diagrams generated by the software are useful.

In general terms, users of Leximancer software submit documents and then omit words they don't want to include in analysis (such as *a* and *the*); then the software performs a keyword count. The body of the text is then broken up into segments, and the concepts that result form a "thesaurus" of words that journey through the text document together.

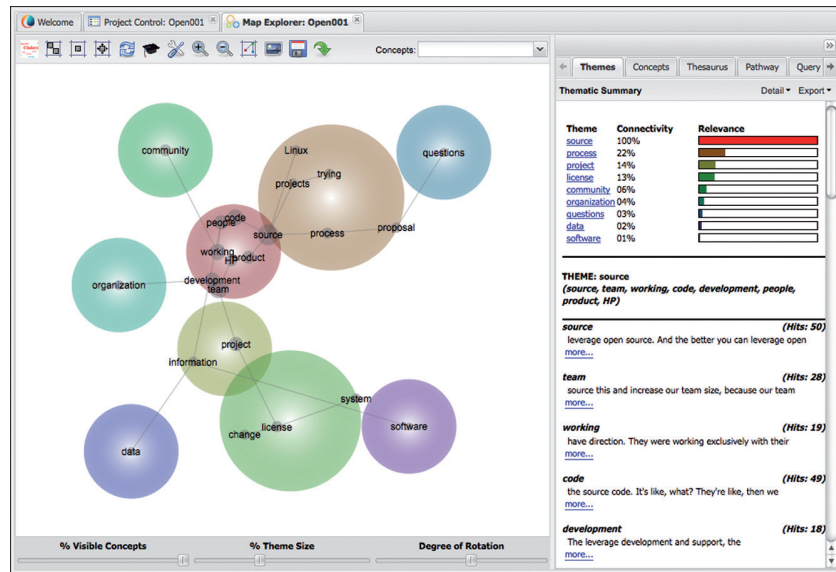
Figure 5.7 shows a concept map generated with some interview data from the Open Source Communities project, creating a different visual display of the text from interviews. As mentioned previously, submitting your documents to the software generates a concept map. Notice how different the data appear, with the user's thesaurus words, such as *Linux*, *projects*, *trying*, and *process* conceptually hanging together and the term *community* separate but related to those concepts through *people*.

Figure 5.8 shows the ranked concepts for the Open Source Communities project using Leximancer. This ranked bar chart format shows the most prominent concepts within the designated category. Notice the prominence of the relationship between the category "community" and the use of the word *development*. The user can visually see the prominent relationship between the category "license" and use of the word *project*.

Your expertise in designing an interface as well as the system is especially critical here. Part of what you must do is demonstrate to the organization the value of using analysis of unstructured, qualitative data to identify and predict future trends in customer, vendor, and supplier behavior. Another important element is to design the human activities surrounding the use of text analytics software. Incorporating the tools is not enough. Remember that you are not just converting qualitative data to quantitative data. Instead, you are developing systems that include text analytics software as a useful means to an end. Include guidelines and suggestions for how decision makers should view the qualitative results, how they might interpret the qualitative results, and how the evidence from those results can be used to make well-reasoned, logically based recommendations, which in turn can be acted on to improve the organization.

FIGURE 5.7

Concept map showing prominence and relationships of concepts in Open Source Communities project interview data.



Observing a Decision Maker's Behavior

Observing decision makers—their physical environment and their interaction with their physical, ergonomic environment—is an important, unobtrusive method a systems analyst may use to gain insight into what is actually done, not just what is documented or explained. In addition, through observation of the decision maker, an analyst can see firsthand the relationships that exist between decision makers and other members of the organization. Observation of decision makers' interactions with technologies also reveals important clues regarding HCI concerns, such as how well the system fits with the user.

Category: community				Category: license			
Concept	Rel Freq (%)	Strength (%)	Prominence	Concept	Rel Freq (%)	Strength (%)	Prominence
development	2	5	4.1	project	11	4	8.6
information	2	5	4.1	system	5	3	7.1
source	4	4	2.9	source	5	2	3.7
system	< 1	< 1	0.0	team	< 1	< 1	0.0
project	< 1	< 1	0.0	organization	< 1	< 1	0.0
team	< 1	< 1	0.0	development	< 1	< 1	0.0
organization	< 1	< 1	0.0	product	< 1	< 1	0.0
product	< 1	< 1	0.0	information	< 1	< 1	0.0

FIGURE 5.8

Ranked concepts for categories overview for the Open Source Communities project using the Leximancer Insight Dashboard. (Two screenshots from Leximancer software. Copyright © by Leximancer Pty Ltd. Reprinted with permission.)

Observing a Typical Manager's Decision-Making Activities

Managers' workdays have been described as a series of interruptions punctuated by short bursts of work. In other words, pinning down what a manager "does" is a slippery proposition, even under the best of circumstances. A systems analyst can grasp how managers characterize their work by using interactive interviews and questionnaires. Observation, however, allows the analyst to see firsthand how managers gather, process, share, and use information and technology to get work done.

Although it is possible to describe and document how managers make decisions using boxes and arrows, we are primarily describing humans and their activities. Therefore, we suggest that systems analysts use a more humanistic approach to describe what managers do. This method is called the analyst's *playscript*. With this technique, the "actor" is the decision maker who is observed "acting" or making decisions. When you set up a playscript, you list the actor in the left-hand column and all of his or her actions in the right-hand column, as shown in Figure 5.9. All activities are recorded with action verbs; for example, a decision maker would be described as "talking," "sampling," "corresponding," and "deciding."

Using a playscript is an organized and systematic approach that demands that the analyst be able to understand and articulate the action taken by each observed decision maker. This approach eventually assists the systems analyst in determining what information is required for

Playscript Analysis	Company: Solid Steel Shelving Analyst: L. Bracket	Scenario: Quality Assurance Date: 1/3/2018
<u>Decision Maker (Actor)</u>	<u>Information-Related Activity (Script)</u>	
Quality Assurance Manager	Asks shop floor supervisor for the day's production report	
Shop Floor Supervisor	Prints out daily computerized production report	
	Discusses recurring problems in production runs with quality assurance (QA) manager	
Quality Assurance Manager	Reads production report	
	Compares current report with other reports from the same week	
	Inputs data from daily production run into QA model on computer	
	Observes onscreen results of QA model	
	Calls steel suppliers to discuss deviations from quality standards	
Shop Floor Supervisor	Attends meeting on new quality specifications with quality assurance manager and vice president of production	
Quality Assurance Manager	Drafts letter to inform suppliers on new quality specifications agreed on in meeting	
	Sends draft to vice president via email	
Vice President of Production	Reads drafted letter	
	Returns corrections and comments via email	
Quality Assurance Manager	Reads corrected letter on email	
	Rewrites letter to reflect changes	

FIGURE 5.9

A sample page from an analyst's playscript describing decision making.

major or frequent decisions made by the observed people. For instance, from the quality assurance manager example in the playscript, it is clear that even though this decision maker is on the middle management level, he or she requires a fair amount of external information to perform the required activities of this specific job.

Observing the Physical Environment

Observing the activities of decision makers is just one way to assess their information requirements. Observing the physical environment where decision makers work also reveals much about their human information requirements. Most often, such observation means systematically examining the offices of decision makers because offices constitute their primary workplace. Decision makers influence, and are in turn influenced by, their physical environments and by their interactions with the technology that take place there. Many HCI concerns can be identified through structured observation and confirmed using other techniques, such as interviews or questionnaires.

Structured Observation of the Environment (STROBE)

Film critics sometimes use a structured form of criticism called *mise-en-scène* analysis to systematically assess what is in a single shot of the film. They look at editing, camera angle, set decor, and the actors and their costumes to find out whether they are shaping the meaning of the film as intended by the director. Sometimes the film's *mise-en-scène* will contradict what is said in the dialogue. For information requirements analysis, a systems analyst can take on a role similar to that of the film critic. It often is possible to observe the particulars of the surroundings that will confirm or negate the organizational narrative (also called *stories* or *dialogue*) that is found through interviews or questionnaires.

*ST*Structured *OB*serva**tion** of the Environment is referred to as STROBE. Successful application of STROBE requires that an analyst explicitly observe seven concrete elements commonly found in offices. The seven observable elements and some key questions that may arise are listed in Figure 5.10. These elements can reveal much about the way a decision maker gathers, processes, stores, and shares information, as well as information about the decision maker's credibility in the workplace.

FIGURE 5.10

Seven concrete observable elements of STROBE and examples of questions an analyst may want to ask.

Observable Element	Questions an Analyst Might Investigate
Office location	Who has the corner office? Are the key decision makers dispersed over separate floors?
Desk placement	Does the placement of the desk encourage communication? Does the placement demonstrate power?
Stationary equipment	Does the decision maker prefer to gather and store information personally? Is the storage area large or small?
Props	Is there evidence that the decision maker uses a PC, smartphone, or tablet computer in the office?
External information sources	Does the decision maker get much information from external sources such as trade journals or the Web?
Office lighting and color	Is the lighting set up to do detailed work or more appropriate for casual communication? Are the colors warm and inviting?
Clothing worn by decision makers	Does the decision maker show authority by wearing conservative suits? Are employees required to wear uniforms?

Characteristics of Decision Makers	Corresponding Elements in the Physical Environment
Gathers information informally	Warm, incandescent lighting and colors
Seeks extraorganization information	Trade journals present in office
Processes data personally	PCs, or tablet computers present in office
Stores information personally	Equipment/files present in office
Exercises power in decision making	Desk placed for power
Exhibits credibility in decision making	Wears authoritative clothing
Shares information with others	Office easily accessible

FIGURE 5.11
A summary of decision maker characteristics that correspond to observable elements in the physical environment.

Anecdotal List with Symbols for Applying STROBE			
Narrative Portrayed by Organization Members	Office Location and Equipment	Office Lighting, Color, and Graphics	Clothing of the Decision Maker
Information is readily flowing on all levels.			
Adams says, "I figure out the percentages myself."			
Vinnie says, "I like to read up on these things."			
Ed says, "The right hand doesn't always know what the left hand is doing."			
Adams says, "Our company doesn't change much."			
The operations staff works all night sometimes.			
Vinnie says, "We do things the way Mr. Adams wants to."			
Julie says, "Stanley doesn't seem to care sometimes."			

Key

Confirm the narrative

Negate or reverse the narrative

Cue to look further

Modify the narrative

Supplement the narrative

FIGURE 5.12
An anecdotal list with symbols for use in applying STROBE.



MAC APPEAL

Collecting data unobtrusively seems easy until one realizes that all the data collected must be organized, stored, and retrieved for analysis. A simple solution to this problem is software called Yojimbo from Bare Bones software. It is inexpensive and easy to use. Just drag the items you want to collect into Yojimbo and search for them when you want to retrieve them.

A more structured approach is to use an application such as DEVONthink Professional Office. The metaphor of an office is a bit ambitious because using the application is more like tossing all sorts of data into a desk drawer and then figuring out how to organize it at a later date. DEVONthink accepts Microsoft Word, Excel, and PowerPoint files as well as anything from iWork. It can keep track of bookmarks and Web pages, images, and PDF files. A built-in OCR reader helps input pages directly.

When it is time to access the information, DEVONthink can help a systems analyst search, classify, and show relationships among items with the help of artificial intelligence (AI). DEVONthink doesn't help an analyst determine the sample size or keep track of errors, but it does help the analyst collect, store, retrieve, use, and share the information gathered.

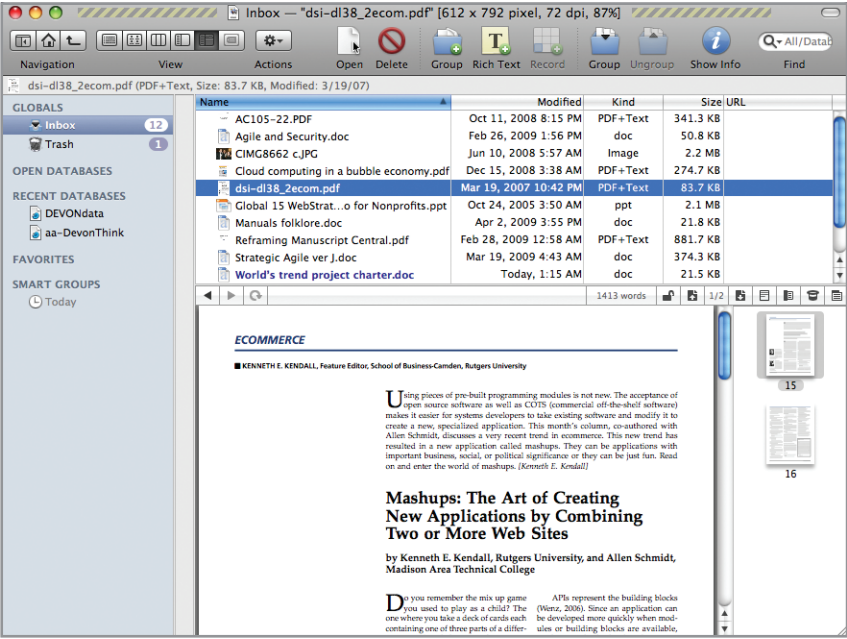


FIGURE 5.MAC

DEVONthink Professional Office from DEVONtechnologies. (Screenshot from DEVONTHINK PRO OFFICE. Copyright © by DEVONtechnologies. Reprinted by permission.)

Through the use of STROBE, a systems analyst can gain a better understanding of how managers gather, process, store, and use information. A summary of the characteristics exhibited by decision makers and the corresponding observable elements is shown in Figure 5.11.

Applying STROBE

One way to implement STROBE is through the use of an anecdotal checklist with meaningful shorthand symbols. This approach to STROBE was useful in ascertaining the information requirements for four key decision makers in a franchise clothing store.

In Figure 5.12, the system analysts used five shorthand symbols to evaluate how observation of the STROBE elements compared with the organization narrative generated through interviews. The five symbols are as follows:

1. A check mark means the narrative is confirmed.
2. An “X” means the narrative is reversed.
3. An oval or eye-shaped symbol serves as a cue for the systems analyst to look further.
4. A square means observation of the STROBE elements modifies the narrative.
5. A circle means the narrative is supplemented by what is observed.

When STROBE is implemented in this manner, the analyst first writes down key organizational themes growing out of interviews. Then he or she observes and records the elements of STROBE. The analyst then compares the narrative and observations and uses one of the five appropriate symbols to characterize the relationship. The analyst thus creates a table that first documents and then aids in the analysis of observations.

Summary

This chapter has covered unobtrusive methods for information gathering, including sampling, investigation of quantitative and qualitative data in current and archived forms, and observation of the decision maker’s activities through the use of the analyst’s playscript, as well as observation of the decision maker’s physical environment through the use of STROBE.

The process of systematically selecting representative elements of a population is called sampling. The purpose of sampling is to select and study documents such as emails, invoices, sales reports, and memos, or perhaps to select and interview, give surveys to, or observe members of the organization. Sampling can reduce cost, speed data gathering, potentially make the study more effective, and possibly reduce the bias in the study.

A systems analyst must follow four steps in designing a good sample: (1) determine the population itself, (2) determine the type of sample, (3) calculate the sample size, and (4) plan the data that need to be collected or described.

The types of samples useful to a systems analyst are convenience samples, purposive samples, simple random samples, and complex random samples. The last type includes the subcategories of systematic sampling and stratified sampling. There are several guidelines to follow when determining sample size.

Systems analysts need to investigate current and archival data and forms that reveal where the organization has been and where its members believe it is going. Text analytics uses software to examine unstructured, qualitative data from blogs, wikis, social media sites, and other online customer interactions to support decision makers in interpreting qualitative material.

Both quantitative and qualitative documents need to be analyzed. Because documents are persuasive messages, it must be recognized that changing them might well change the organization.

Analysts use observation as an information-gathering technique. Through observation, they gain insight into what is actually done as users interact with information technology. One way to describe how decision makers behave is to use an analyst’s playscript that documents each of the major players’ activities.

In addition to observing a decision maker’s behavior, a systems analyst should observe the decision maker’s surroundings for important clues as to how well the system fits the user. One method is STROBE. A systems analyst uses STROBE in the same way that a film critic uses a method called *mise-en-scène* analysis to analyze a shot in a film.

A systems analyst can observe and interpret several concrete elements in the decision maker’s environment. These elements include (1) office location, (2) placement of the decision maker’s desk, (3) stationary office equipment, (4) props such as handheld devices and PCs, (5) external information sources such as trade journals and use of the Web, (6) office lighting and color, and (7) clothing worn by the decision maker. A systems analyst can use STROBE to gain a better understanding of how decision makers actually gather, process, store, and share information in order to get their work done.



HYPERCASE EXPERIENCE 5.2

“We’re proud of our building here in Tennessee. In fact, we used the architectural firm of I. M. Paid to carry the same theme, blending into the local landscape while still reaching out to our clients throughout all the branches. We get lots of people coming through just to admire the building once they catch on to where it is exactly. In fact, by Tennessee standards, we get so many sightseers that it might as well be the pyramids! Well, you can see for yourself as you go through. The East Atrium is my favorite place: plenty of light, a huge skylight overhead. Yet it has always fascinated me that the building and its furnishings might tell a story quite different from the one its occupants tell.”

“Sometimes employees complain that the offices all look the same. The public rooms are spectacular though. Even the canteen is inviting. Most people can’t say that about their cafeterias at work. You’ll notice that we all personalize our offices anyway. So even if the offices were of the ‘cookie cutter’ kind, their occupants’ personalities seem to take over as soon as they have been here a while. What have you seen? Was there anything that surprised you so far?”

HYPERCASE Questions

1. Use STROBE to compare and contrast Mr. Evans’ and Mr. Ketcham’s offices. What sort of conclusion about each person’s use of information technology can you draw from your observations? How compatible do Evans and Ketcham seem in terms of the systems they use? What other clues to their storage, use, and sharing of information can you discover based on your observations of their offices?
2. Carefully examine Kathy Blandford’s office. Use STROBE to confirm, reverse, or negate what you have learned during your interview with her. List anything you found out about Ms. Blandford from observing her office that you did not know from the interview.
3. Carefully examine the contents of the MRE reception area using STROBE. What inferences can you make about the organization? List them. What interview questions would you like to ask based on your observations of the reception area? Make a list of people you would like to interview and the questions you would ask each of them.



FIGURE 5.HC1

There are hidden clues in HyperCase. Use STROBE to discover them.

Keywords and Phrases

analyst's playscript	qualitative documents
business-to-business (B2B) ecommerce	quantitative documents
business-to-consumer (B2C) ecommerce	sample population
cluster sampling	sampling
complex random sample	simple random sample
confidence level	stratified sampling
convenience sample	STROBE
corporate websites	systematic observation
external information sources	systematic sampling
office lighting and color	text analytics
purposive sample	

Review Questions

1. Which two key sampling issues does a systems analyst need to address?
2. List four reasons a systems analyst would want to sample data or select representative people to interview.
3. What are the four steps to follow to design a good sample?
4. List the three approaches to complex random sampling.
5. Define what is meant by *stratification of samples*.
6. List seven steps used by systems analysts to determine sample size under ideal conditions.
7. What is the overriding variable that determines how many people a systems analyst should interview in depth?
8. List the guidelines for a systematic examination of qualitative documents.
9. What are the sources of big data for text analytics that an analyst might use? List them.
10. Company A has 500 employees and Company B has 800 employees. Would the analyst working with Company B need to interview more employees than the analyst working with Company A? Why or why not?
11. List five steps to help an analyst observe a decision maker's typical activities.
12. Explain the importance of a playscript for a systems analyst during observation.
13. Discuss the role of the physical environment of a decision maker in the determination of their HCI requirements.
14. Describe how a systems analyst implements STROBE through an anecdotal checklist and symbols.
15. List the seven concrete elements of a decision maker's physical environment that a systems analyst can observe using STROBE.

Problems

1. You are hired as a systems analyst by an organization that is planning to redesign its website. Consider the following situations and describe the most appropriate sampling method for each of them.
 - a. To gauge employee perception towards website redesign, you post a notice on the intranet portal asking the employees to post their opinions.
 - b. You design a customer survey to find out which website features they wish to see redesigned.
 - c. You seek customer opinions through forms available at three of the company's 20 helpdesks.
 Explain why the answer to each situation would vary.
2. "I see that you have quite a few papers there. What all do you have in there?" asks Betty Kant, head of the MIS task force that is the liaison group between your systems group and Sawder's Furniture Company. You are shuffling a large bundle of papers as you prepare to leave the building.

"Well, I've got some financial statements, production reports from the last six months, and some performance reports that Sharon gave me that cover goals and work performance over the last six months," you reply as some of the papers fall to the floor. "Why do you ask?"

Betty takes the papers from you and puts them on the nearest desk. She answers, "Because you don't need all this junk. You're here to do one thing, and that's talk to us, the users. Bet you can't read one thing in there that'll make a difference."

 - a. The only way to convince Betty of the importance of each document is to tell her what you are looking for in each one. In a paragraph, explain what each kind of document contributes to a systems analyst's understanding of the business.

- b. While you are speaking with Betty, you realize that you actually need other quantitative documents as well. List any you are missing.
3. You've sampled the email messages that have been sent to several middle managers of Sawder's Furniture Company, which ships build-your-own particleboard furniture across the country. Here is one that repeats a message found in several other memos:

To: Sid, Ernie, Carl
 From: Imogene
 Re: Computer/printer supplies
 Date: November 10, 2017

It has come to my attention that I have been waging a war against requests for computer and printer supplies (USBs, toner, paper, and so on) that are all out of proportion to what has been negotiated for in the current budget. Because we're all good soldiers here, I hope you will take whatever our supply sergeant says is standard issue. Please, no "midnight requisitioning" to make up for shortages. Thanks for being GI in this regard; it makes the battle easier for us all.

- a. What metaphor(s) is (are) being used? List the predominant metaphor and other phrases that play on that theme.
- b. If you found repeated evidence of this idea in other email messages, what interpretation would you have? Use a paragraph to explain.
- c. In a paragraph, describe how the people in your systems analysis group can use the information from the email messages to shape their systems project for Sawder's.
- d. In interviews with Sid, Ernie, and Carl, there has been no mention of problems with obtaining enough computer and printer supplies. In a paragraph, discuss why such problems may not come up in interviews and discuss the value of examining email messages and other memos in addition to interviewing.
4. "Here's the main policy manual we've put together over the years for system users," says Al Bookbinder, as he blows the dust off the manual and hands it to you. Al is a document keeper for the systems department of Prechter and Gumbel, a large manufacturer of health and beauty aids. "Everything any user of any part of the system needs to know is in what I call the Blue Book. I mean it's chockablock with policies. It's so big, I'm the only one with a complete copy. It costs too much to reproduce it." You thank Al and take the manual with you. When you read through it, you are astonished at what it contains. Most pages begin with a message such as: "This page supersedes page 23.1 in manual Vol. II. Discard previous inserts; do not use."
 - a. List your observations about the frequency of use of the Blue Book.
 - b. How user friendly are the updates in the manual? Write a sentence explaining your answer.
 - c. Write a paragraph commenting on the wisdom of having all-important policies for all systems users in one book.
 - d. Suggest a solution that incorporates the use of online policy manuals for some users.
 - e. Write a paragraph about how text analytics could help the organization do a review of the content of employee policies posted online. Write a second paragraph stating whether you would recommend this approach and why or why not.
5. "I think I'll be able to remember most everything he does," says Ceci Awll. Ceci is about to interview Biff Welldon, vice president of strategic planning of Steak Your Claim, a steak restaurant chain with 130 locations. "I mean, I've got a good memory. I think it's much more important to listen to what he says than to observe what he does anyway." As one of your systems analysis team members, Ceci has been talking with you about the desirability of writing down her observations of Biff's office and activities during the interview.
 - a. In a paragraph, persuade Ceci that listening is not enough in interviews and that observing and recording those observations are also important.
 - b. Ceci seems to have accepted your idea that observation is important but still doesn't know what to observe. Make a list of items and behaviors to observe, and in a sentence beside each behavior, indicate what information Ceci should hope to gain through observation of it.
6. A systems analyst is part of an information systems project for a firm, and while gathering information, she gets a chance to observe its physical environment. She finds that this large firm has assigned offices for representatives and executives on different floors, and there are vast spaces between the staff and the executive offices demarcated by a large door that displays a strict schedule for meeting the executives. Generally, each of these offices has two or three chairs for visitors in addition to the usual file storage cabinets, CD cabinets, etc., but almost all of them seem to be in disuse. All the rooms have PCs, but the analyst observes a lack of proper electrical equipment and wiring that connects these machines to other parts of the office and to the Internet. In addition, there is no stationary found at the desks in these offices, though it is scattered across the desks of the other staff in abundance. All the rooms are adequately lit, with some executives using brighter lighting than others. Some of the leaders are casually dressed, while most of them are in formal attire. During information gathering, a lot of employees refrained from discussing their relationships with the management. The

executives often reported using external sources of information to take non-routine decisions. They also said that they overlooked all the information on personal matters passed around in the office.

- a. What are the organizational narratives or themes that emerge from this scenario?
- b. List the elements of STROBE that the systems analyst has reported.
- c. For each of the elements listed, prepare a summary of the characteristics of executives that correspond to the observable elements in the physical environment.
- d. Construct a table first by listing the organizational themes or narratives emerging from the scenario. Then, for each narrative, using shorthand symbols from the anecdotal list, indicate the relationship between the narrative and the observation, elucidating whether the narrative confirms, reverses, needs further investigation, needs modification, or supplements the observable elements of the environment.
- e. The analyst also heard executives often stating that employees get open access to their offices to discuss their ideas, opinions, and problems. With your interpretation of the STROBE elements observed by the analyst, would you agree with the executives' statement? Why or why not?

Group Projects

1. Assume that your group will serve as a systems analysis and design team for a project designed to computerize or enhance the computerization of all business aspects of a 15-year-old national U.S. trucking firm called Maverick Transport. Maverick is a less-than-a-truckload (LTL) carrier. The people in management work from the philosophy of just in time (JIT), in which they have created a partnership that includes the shipper, the receiver, and the carrier (Maverick Transport) for the purpose of transporting and delivering the materials required just in time for their use on the production line. Maverick maintains 626 tractors for hauling freight and has 45,000 square feet of warehouse space and 21,000 square feet of office space.
 - a. Along with your group members, develop a list of sources of archival data that should be checked when analyzing the information requirements of Maverick.
 - b. When this list is complete, devise a sampling scheme that would permit your group to get a clear picture of the company without having to read each document generated in its 15-year history.
2. Arrange to visit a local organization that is expanding or otherwise enhancing its information systems. To allow your group to practice the various observation methods described in this chapter, assign either of these two methods to each team member: developing the analyst's (1) playscript or (2) using STROBE. Many of these strategies can be employed during one-on-one interviews, whereas some require formal organizational meetings. To accomplish several objectives during your visit to the organization, schedule your visit at a time that allows all team members to try their assigned method of observation. Using multiple methods such as interviewing and observation (often simultaneously) is the only cost-effective way to get a true, timely picture of the organization's information requirements. Alternatively, if you are working on this problem as a face-to-face group in a school or university setting, arrange to visit one of your school or university's departments during working hours and follow the instructions given above.
3. The members of your group should meet and discuss their findings after completing Group Project 2. Were there any surprises? Did the information garnered through observation confirm, reverse, or negate what was learned in interviews? Were any of the findings from the observational methods in direct conflict with each other? Work with your group to develop a list of ways to address any puzzling information (for example, by doing follow-up interviews). Write down the list and share it with the entire group.
4. With your group, research software solutions that can help analysts keep track of a variety of data such as observations, surveys, and so on. (Hint: One app is Devonthink Professional Office highlighted in Mac Appeal in this chapter). Allow each group member to work on his or her own to find a suitable package, noting advantages and drawbacks. Take about 20 minutes to do so. Once your research is completed, collaborate on creating a table of all the software found by team members. List each piece of software and also give advantages and drawbacks of each.

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